

exmp: let F be the set of all rational numbers / fractions.

$$F = \left\{ \frac{x}{y} : x \in \mathbb{Z}, y \in \mathbb{Z}, y \neq 0 \right\}$$

define R on F : $\frac{x_1}{y_1} R \frac{x_2}{y_2} : x_1 y_2 = x_2 y_1$

Prove R is equivalence relation, identify equ. classe.

Reflexive: $\forall \frac{x_1}{y_1} \in F : \frac{x_1}{y_1} R \frac{x_1}{y_1}$

$$x_1 y_1 = x_1 y_1 \Rightarrow \frac{x_1}{y_1} R \frac{x_1}{y_1} \quad \checkmark$$

Symmetric: $\forall \frac{x_1}{y_1}, \frac{x_2}{y_2} \in F : \frac{x_1}{y_1} R \frac{x_2}{y_2} \stackrel{?}{\Rightarrow} \frac{x_2}{y_2} R \frac{x_1}{y_1}$

$$x_1 y_2 = x_2 y_1 \Rightarrow x_2 y_1 = x_1 y_2 \Rightarrow \frac{x_2}{y_2} = \frac{x_1}{y_1} \quad \checkmark$$

transitive: $\forall \frac{x_1}{y_1}, \frac{x_2}{y_2}, \frac{x_3}{y_3} \in F : \frac{x_1}{y_1} R \frac{x_2}{y_2} \wedge \frac{x_2}{y_2} R \frac{x_3}{y_3} \stackrel{?}{\Rightarrow} \frac{x_1}{y_1} R \frac{x_3}{y_3}$

$$x_1 y_2 = x_2 y_1$$

①

$$x_2 y_3 = x_3 y_2$$

②

$$\textcircled{1} \quad x_1 y_2 = x_2 y_1 \xrightarrow{\cdot y_3} x_1 y_2 y_3 = x_2 y_1 y_3 \xrightarrow{\textcircled{2}} x_1 y_2 y_3 = x_3 y_2 y_1$$

$$\xrightarrow{\div y_2} x_1 y_3 = x_3 y_1$$

$$\rightarrow \frac{x_1}{y_1} R \frac{x_3}{y_3} \quad \checkmark$$

$\mathbb{R}$ partitioning F into some equivalence classes:

$$[\frac{1}{2}] = \{ \dots, -\frac{1}{2}, \frac{1}{2}, \frac{2}{4}, \frac{3}{6}, \frac{4}{8}, \dots, \frac{k}{2k}, \dots \}$$

$$[\frac{5}{4}] = \{ \dots, -\frac{5}{4}, \frac{5}{4}, \frac{10}{8}, \frac{15}{12}, \dots, \frac{5k}{4k}, \dots \}$$

$\vdots$

$$[3] = \{ \dots, -\frac{6}{2}, \frac{3}{1}, \frac{6}{2}, \frac{9}{3}, \dots, \frac{3k}{k}, \dots \}$$

$$[-3] = \{ \dots, \frac{-3}{1}, \frac{3}{-1}, \frac{-6}{2}, \frac{6}{-2}, \dots \}$$

$\vdots$

$\vdots$

$\vdots$

+ Midterm

- Wed, Feb 25th

- lecture room (10-11:20)

- 30% of final mark

- covers all lectures 1-12 (including today lecture)

- open book (any document, no electronics)

Graph theory: are used to make ~~computer~~ networks, electrical networks, roads, wireless networks, AI, graphics, data science, ...

$G(V, E)$: consists of a set V of vertices and a set E of edges of the graph, where each edge connects two vertices.

example: $G = (V, E)$

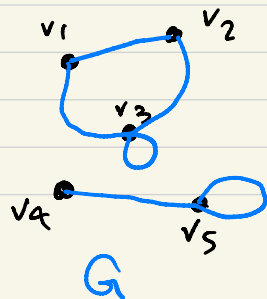
$$V = \{\underline{v_1}, \underline{v_2}, \underline{v_3}, \underline{v_4}, \underline{v_5}\} \quad |V| = 5$$

$$E = \{\underline{(v_1, v_2)}, \underline{(v_1, v_3)}, \underline{(v_2, v_3)}, \underline{(v_3, v_3)}, \underline{(v_4, v_5)}, \underline{(v_5, v_5)}\} \quad |E| = 6$$

We usually use a graphical representation (drawing) of graph.

vertex $\rightarrow$ represented by a point

edge $\rightarrow$ represented by a curve (segment) between vertices.



undirected edges: $(u, v) = (v, u)$
↳ pair (unorderd)



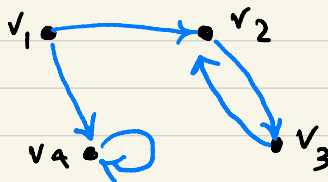
directed edges: $\langle u, v \rangle \neq \langle v, u \rangle$
↳ ordered pair



directed graph: a graph with directed edges

$$G = (V, E)$$

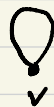
$$V = \{v_1, v_2, v_3, v_4\}$$



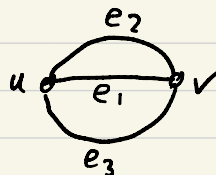
$$E = \{ \langle v_1, v_2 \rangle, \langle v_2, v_3 \rangle, \langle v_3, v_2 \rangle, \langle v_1, v_4 \rangle, \langle v_4, v_4 \rangle \}$$

In this chapter we focus on undirected graphs.

- loop: an edge from a vertex to itself (v, v)



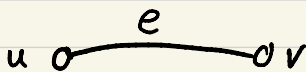
- multi-edge (parallel edges): multiple edges between two vertices



- simple graph: a graph with no loop and no multi-edges.

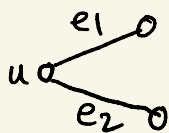
focus of this chapter is simple graphs.

$$e = (u, v)$$



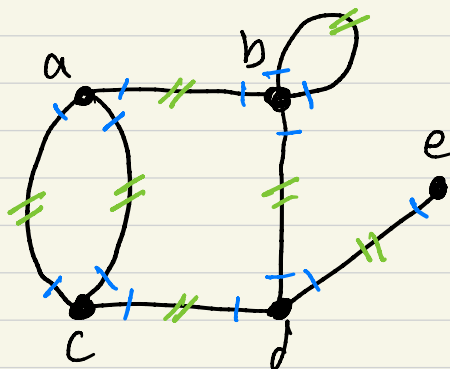
u and v are neighbors / adjacent

e is incident to u and v



e_1 and e_2 are neighbors/adjacent.

degree of a vertex v , $d(v)$: # edges incident to v
(loops counted twice)



$$d(a) = 3$$

$$d(c) = 3$$

$$d(d) = 3$$

$$d(e) = 1$$

$$d(b) = 4$$

$$\sum_{u \in V} d(u) =$$

$$d(a) + d(b) + d(c) + d(d) + d(e)$$

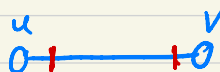
$$= 3 + 4 + 3 + 3 + 1$$

$$= 14$$

$$|E| = 7$$

Theorem: In any graph the sum of degrees is twice the number of edges.

$$\sum_{u \in V} d(u) = 2|E| \quad \text{even}$$

Proof: any edge (u, v)  contributes 1 to $d(u)$

and 1 to $d(v)$. Thus any edge contributes 2 to the

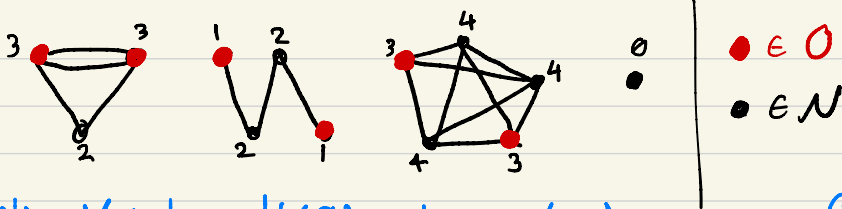
total degree. therefore total degree is twice # of edges.

sum of degrees

$$\sum_{u \in V} d(u)$$

$$2|E|$$

Theorem: The number of odd-degree vertices is even.



Proof: Partition V into odd (O) and even (N)
 $O = \{a, c, d, e\}$
 $N = \{b\}$

$$\sum_{u \in V} d(u) = \sum_{u \in O} d(u) + \underbrace{\sum_{u \in N} d(u)}_{\text{even}} = \underbrace{2|E|}_{\text{an even number}}$$

↑
Previous thm.

because it is sum of even degrees

→ this must be even

$$\sum_{u \in O} d(u) = \text{even}$$

↑
odd

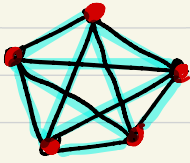
from this we conclude that the number of elements in O must be even. $|O| = \text{even}$. therefore

$$O = \{1, 3, 5\} \Rightarrow 1+3+5=9 \text{ not even}$$

$$O = \{1, 3, 5, 3\} \Rightarrow 1+3+5+3=12 \text{ even}$$

the number of odd-degree vertices is even $\square$

Theorem: Let $G=(V,E)$ be a simple graph. Then $|E| \leq \frac{|V|(|V|-1)}{2}$.

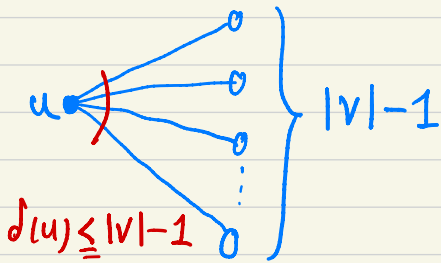


$$|V|=5$$

$$|E|=10$$

$$|E| \leq \frac{5(5-1)}{2} = 10$$

Proof: take any vertex u . The degree of u is at most $|V|-1$



- We have $|V|$ vertices

- degree of each vertex $\leq |V|-1$

$\Rightarrow$ total degree $\leq |V|(|V|-1)$

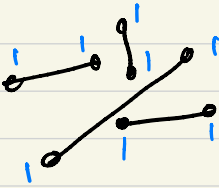
first theorem

$$2|E| = \sum_{u \in V} d(u) \leq |V|(|V|-1)$$

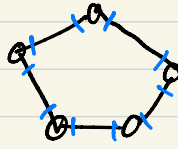
$$\Rightarrow 2|E| \leq |V|(|V|-1)$$

$$\Rightarrow |E| \leq \frac{|V|(|V|-1)}{2} \quad \square$$

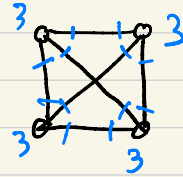
d-regular graph: a graph where the degree of every vertex is d .



1-regular
(matchings)



2-regular
(cycle)



3-regular

H.W. Prove that if G is a d -regular graph with n vertices

then $|E| = \frac{nd}{2}$.

Midterm (up to here).